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LAB: CAPABILITIES OF MICROSOFT SECURITY SOLUTIONS

1.1 INTRODUCTION

I started the module with the required lab setup by pressing Ctrl + Alt + Delete to bring up the logon page. I then logged in as admin using the provided password and clicked on Next to proceed to the lab

1.2 LAB 1:SET UP OF MICROSOFT 365 TENANT

LAB SCENARIO

This setup lab consisted of enabling the Microsoft Audit Log and file monitoring capabilities in the Microsoft 365 tenant.

Step 1

I opened Microsoft Edge and entered the url provided.

Step 2

I signed in with the admin credentials for the Microsoft 365 tenant provided by the authorized lab hoster (ALH)

Step 3

From the left navigation pane of the Microsoft 365 admin center, I selected Show all.

Step 4

Under Admin centers, I selected Security and a new browser page opened to the welcome page of Microsoft Defender.

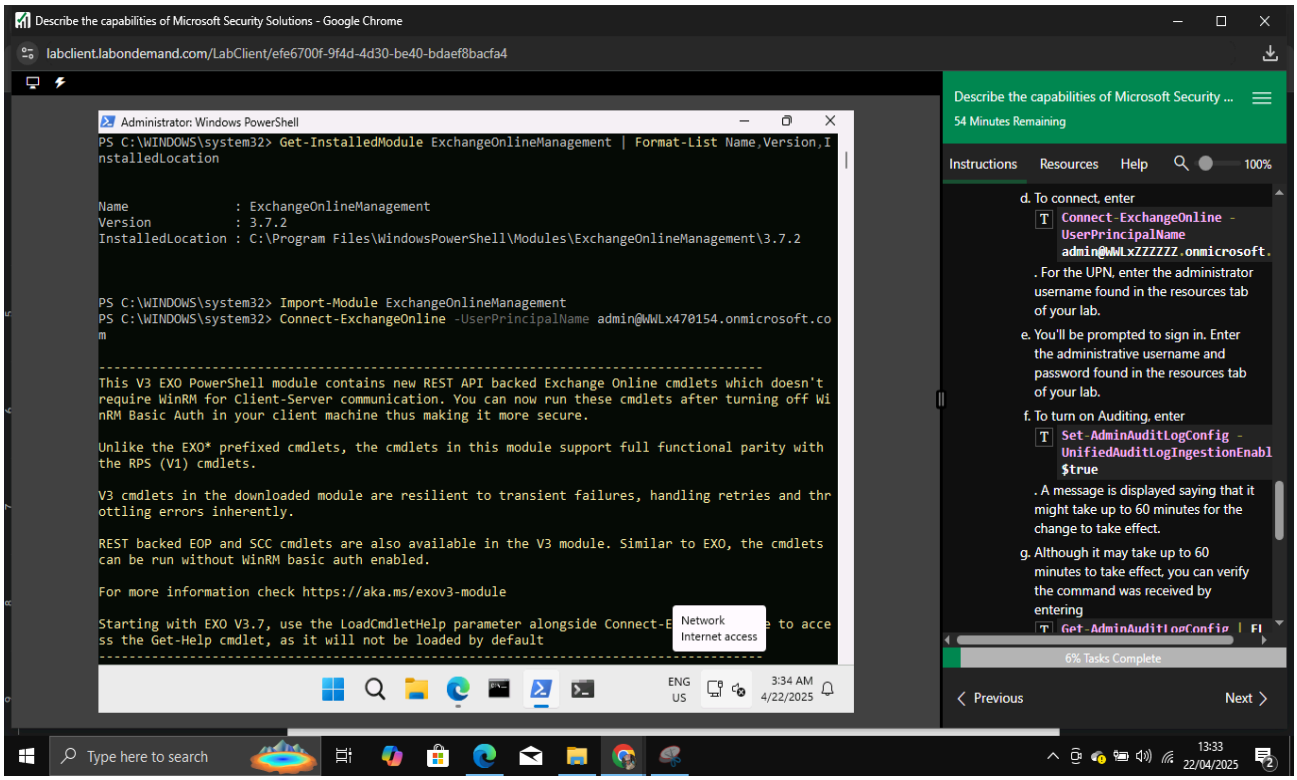
Step 5

In the left navigation panel, I scrolled down and expanded on System. From the expanded list, I selected Audit.

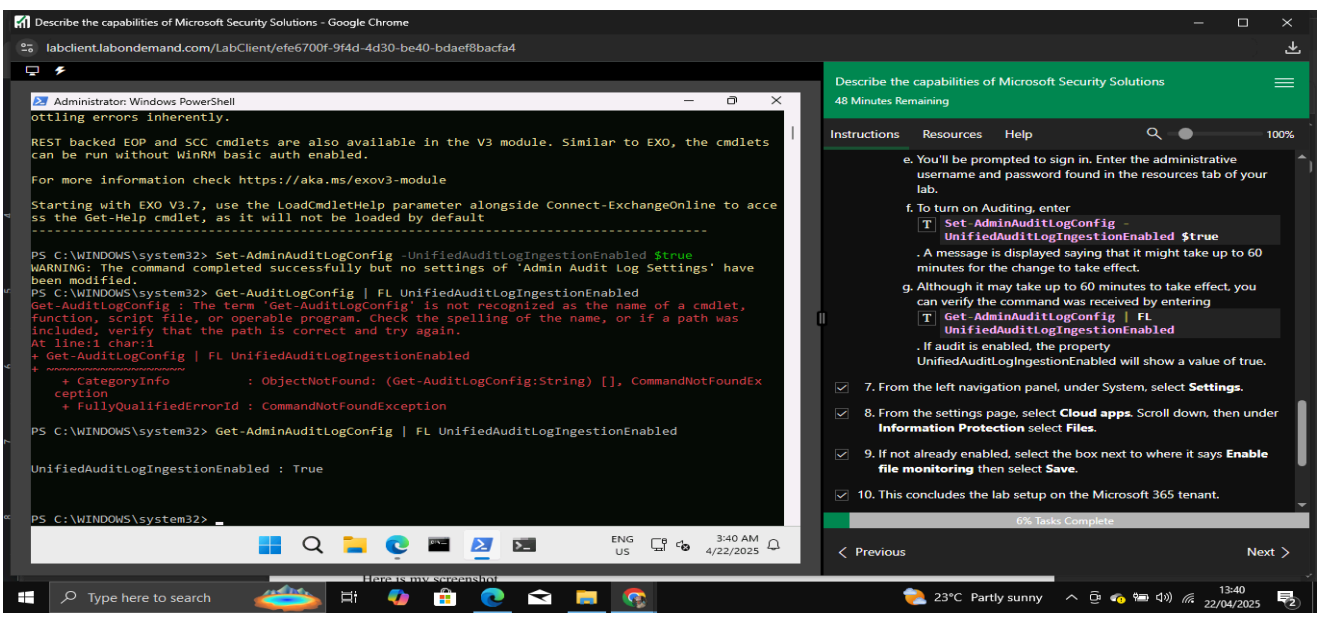
Step 6

In this step,auditing was already enabled.I went ahead to confirm that the Exchange Online PowerShell module is installed on the computer, loaded and connected to it.

Here is my screenshot



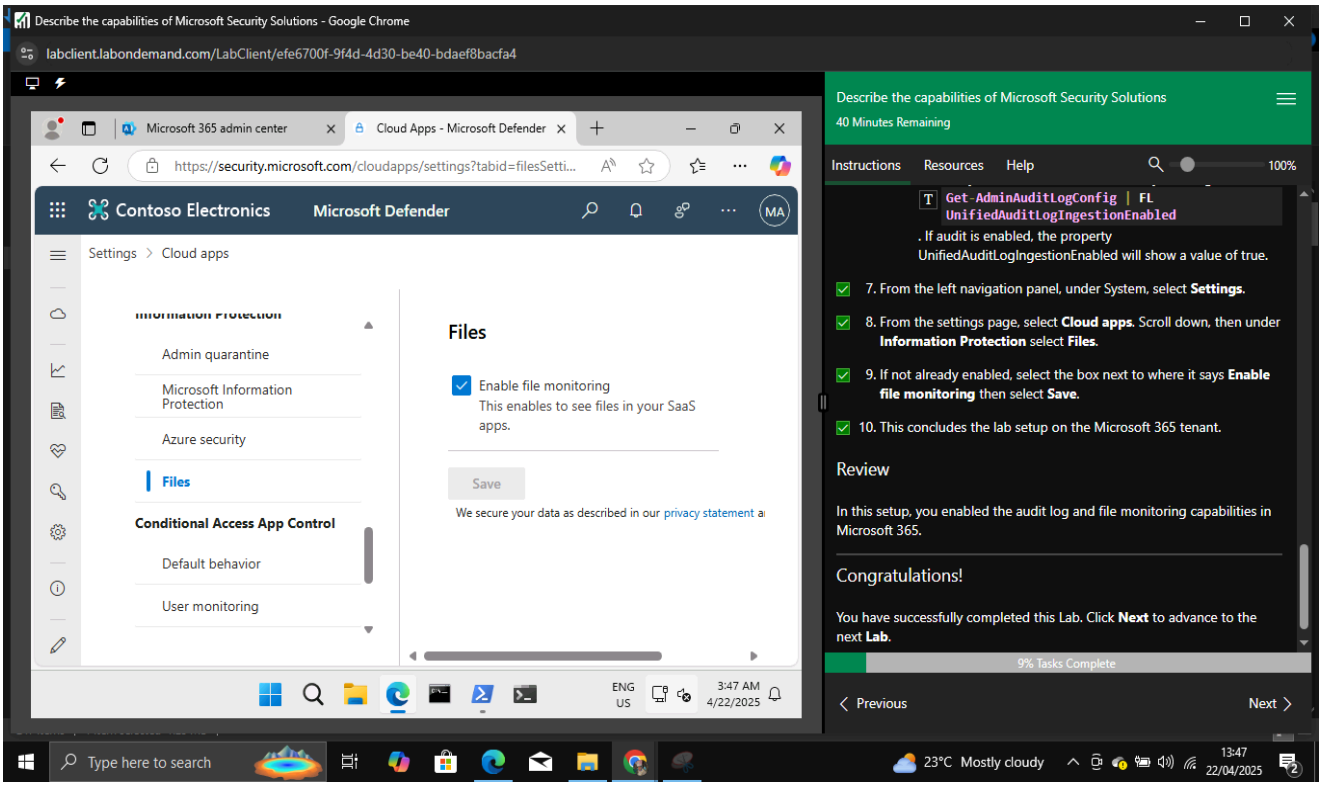
I also verified that audit is enabled, the property UnifiedAuditLogIngestionEnabled shows a value of true. Here is my screenshot



Step 7,8,9 and 10

From the left navigation panel under system, I selected settings then cloud apps, under information protection,I selected files, enabled file monitoring and saved.

Here is my screenshot



1.3 LAB 2:EXPLORE AZURE NETWORK SECURITY GROUPS (NSGs)

LAB SCENARIO

In this lab.I explored the function of network security groups in Azure by creating a network security group (NSG) and assigning the NSG to the interface of a pre-existing virtual machine (VM).Once configured I'd observe the default inbound and outbound rules, create new rules, and test those rules.

TASK 1

In this task,I was required to view some of the parameters associated with the VM that was created for use with this lab.

Step 1

I opened Microsoft Edge and entered the url provided

Step 2

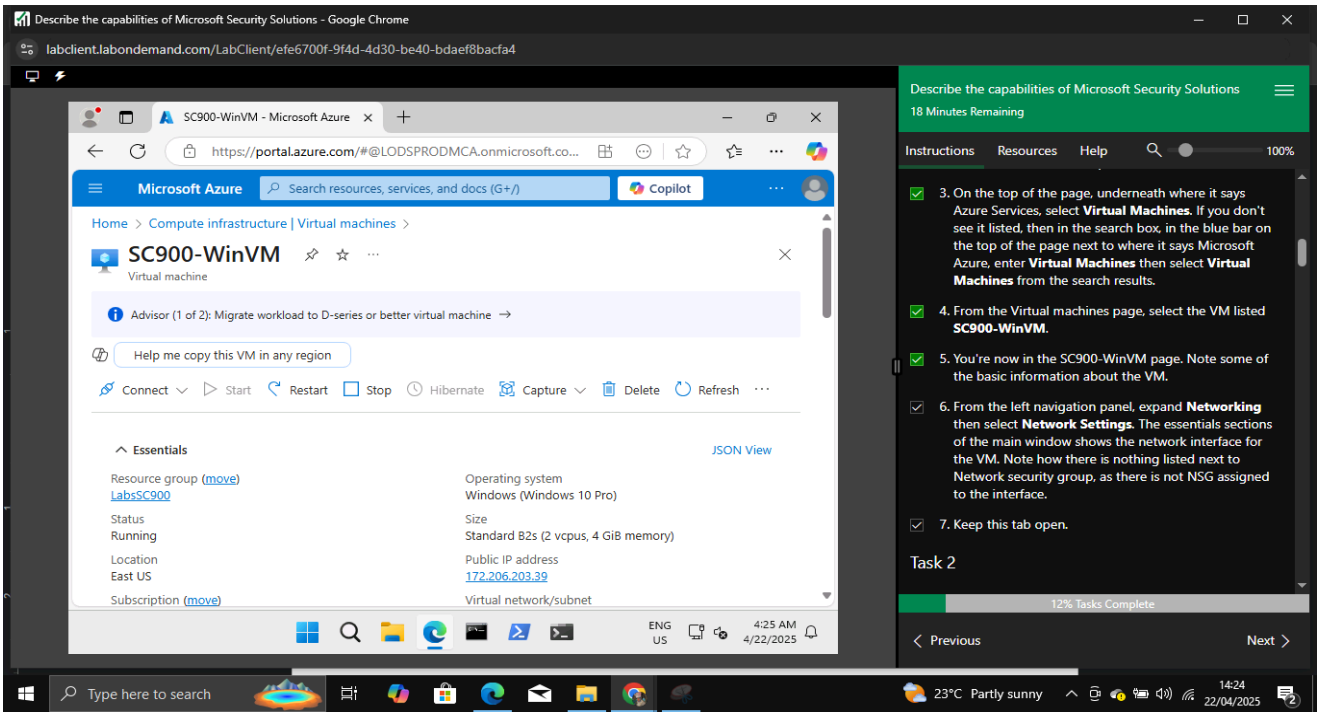
I signed in with my Azure portal credentials provided.

Step 3 and 4

On the Azure services,I selected virtual machines and SC900-WinVM

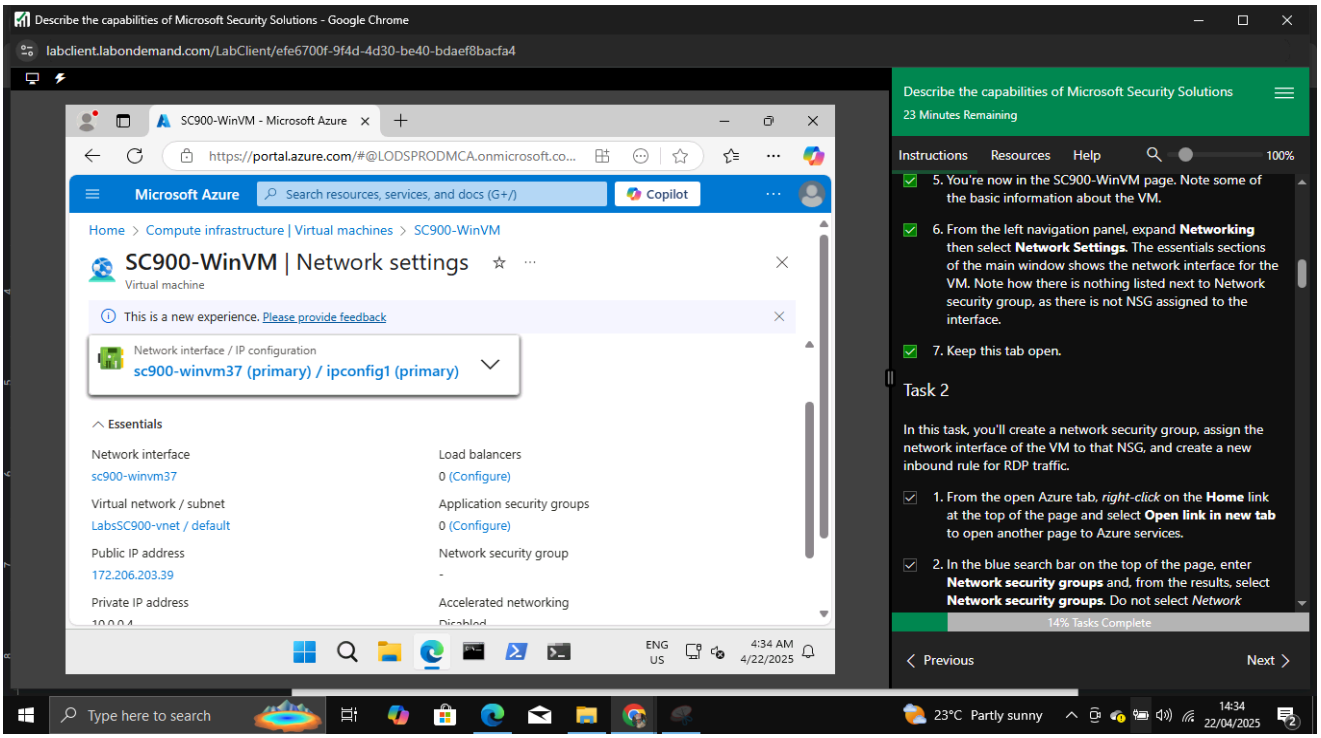
Step 5

I was taken to the SC900-WinVM page and noted the basic information about the VM.
Here is my screenshot



Step 6

From the left navigation panel,I expanded Networking then selected Network Settings.The essentials sections of the main window showed the network interface for the VM and there was no NSG assigned to the interface.
Here is my screenshot



Step 7

I left the tab in step 6 above open.

TASK 2

In this task, I was to create a network security group, assign the network interface of the VM to that NSG, and create a new inbound rule for RDP traffic.

Step 1

From the open Azure tab, I right-clicked on the Home link at the top of the page and selected Open link in new tab to open another page to Azure service

Step 2

I entered Network security groups in the search bar and selected it.

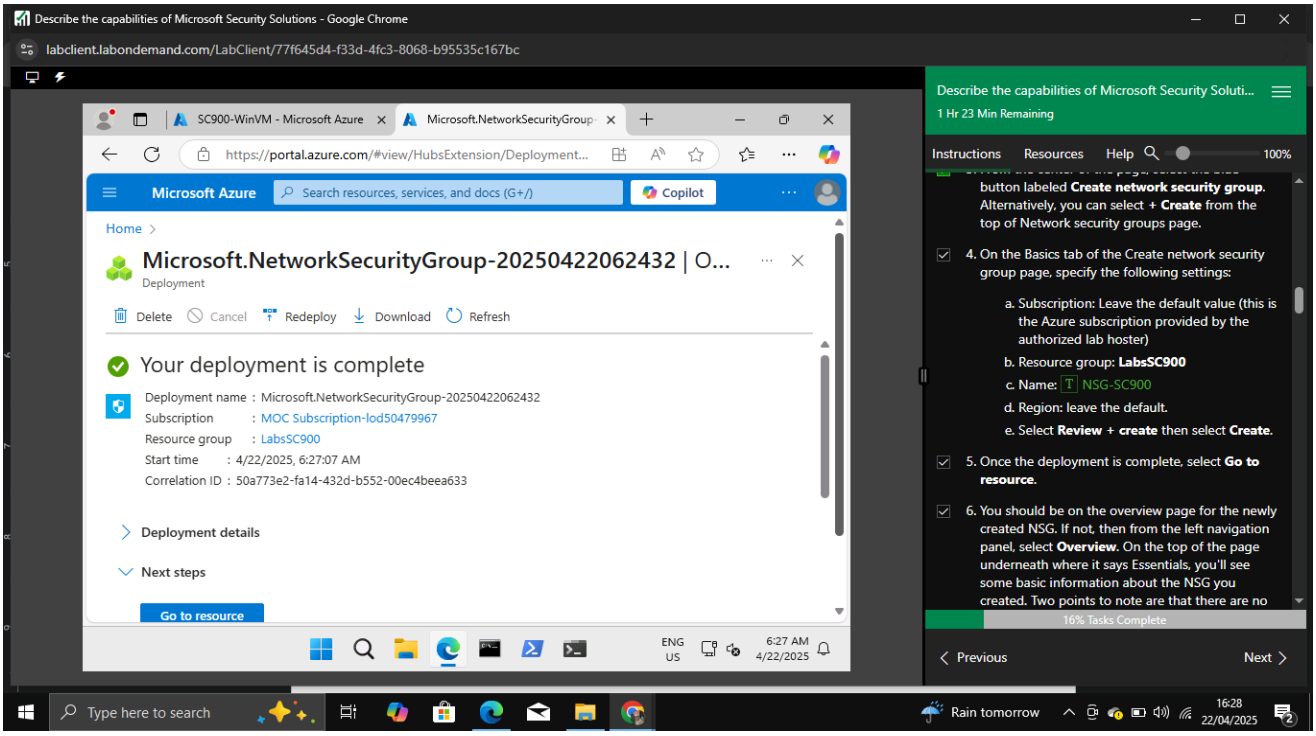
Step 3

From the center of the page,I selected the blue button labeled Create network security group.

Step 4

I created a network security group page, using specified settings

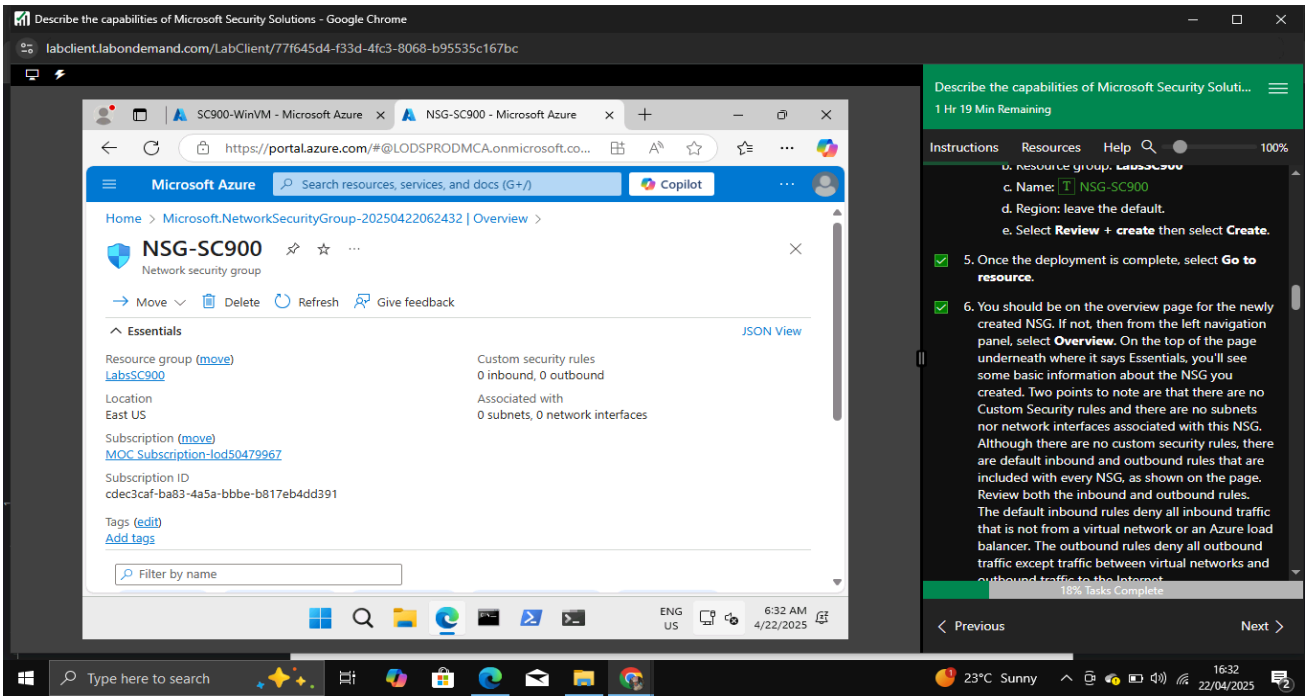
Here is my screenshot



Step 5 and 6

I selected " Go to resource",and was taken to the overview page for the newly created NSG.

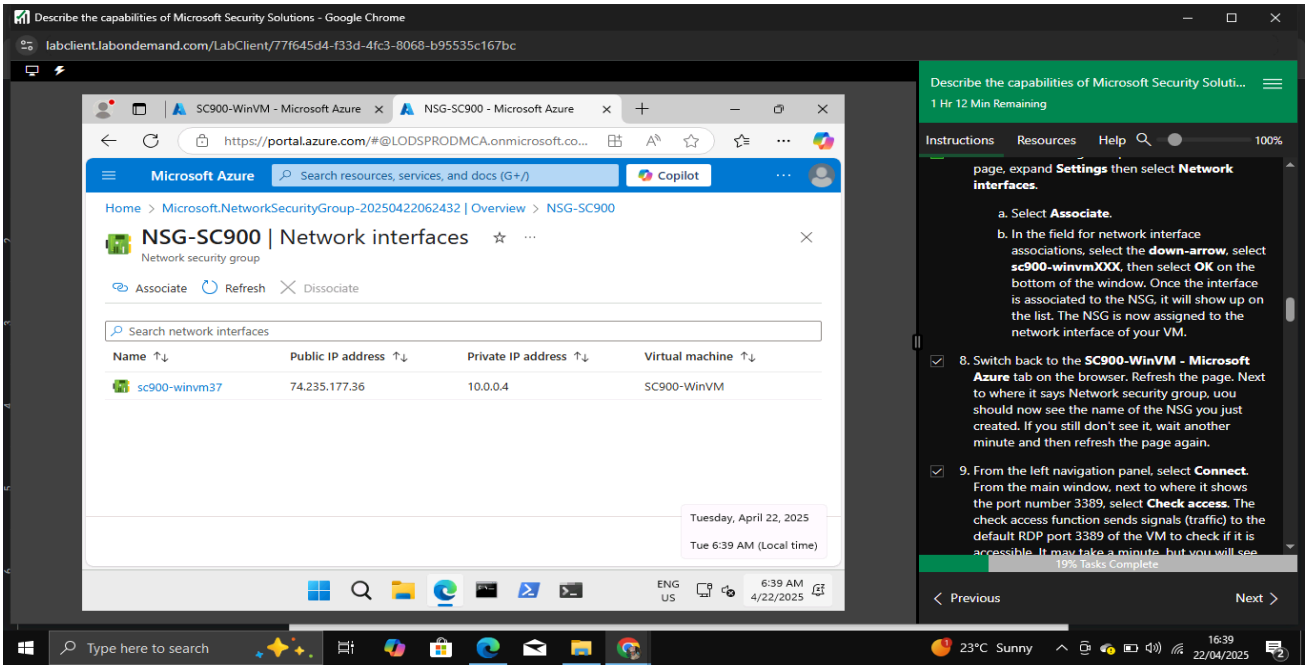
Here is my screenshot



Step 7

From the left navigation pane on the NSG-SC900 page, I expanded Settings then selected Network interfaces,associate,and in the field for network interface associations, I selected the down-arrow,to select the vm name , then OK on the bottom of the window. Once the interface is associated with the NSG, it would show up on the list. The NSG is now assigned to the network interface of my VM.

Here is my screenshot



Step 8

I switched back to my previous tab,refreshed the page and next to where it says Network security group, I saw the name of the NSG I just created.

Step 9

From the left navigation panel,I selected Connect. From the main window, next to where it shows the port number 3389, selected Check access. The check access function sends signals (traffic) to the default RDP port 3389 of the VM to check if it is accessible.

Step 10

I Switched back to the NSG-SC900 - Microsoft Azure tab on the browser

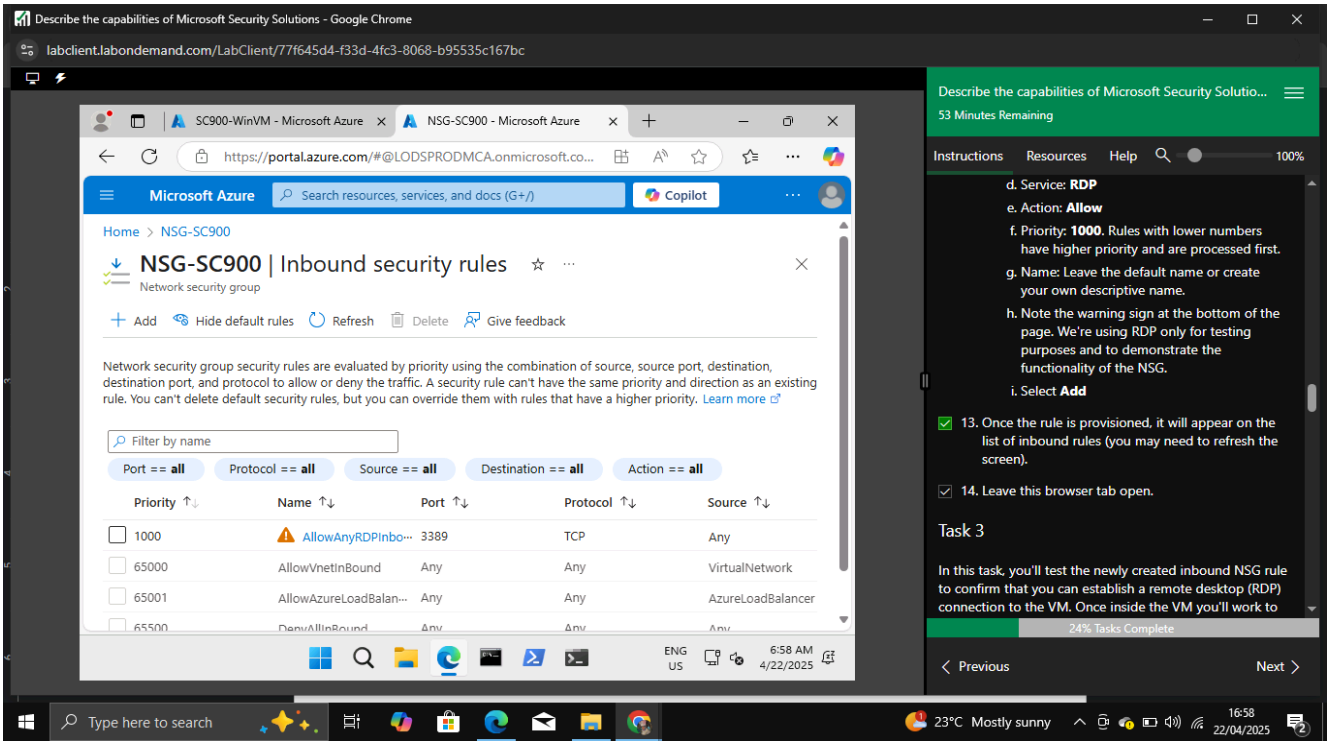
Step 11

From the left navigation pane, I selected Inbound security rules. The default inbound rules deny all inbound traffic that is not from a virtual network or an Azure load balancer so I needed to set up a rule to allow inbound RDP traffic (traffic on port 3389).

Step 12

From the top of the page,I selected Add and specified the settings provided and once the rule was provisioned it appeared on the list of inbound rules

Here is my screenshot



Step 13

I left the browser tab in the previous step open

TASK 3

In this task,I tested the newly created inbound NSG rule to confirm that I can establish a remote desktop (RDP) connection to the VM. Once inside the VM I checked outbound connectivity to the Internet from the VM.

Step 1

I opened the SC900-WinVM – Microsoft Azure Tab on my browser

Step 2

I then selected connect from the left navigation panel.

Step 3

I selected check access to verify that the port is set to 3389 and the status shows "Accessible”

Step 4

I connected directly to the VM by clicking Select in the box that says Native RDP.From the Native RDP window that opened,I selected the Download RDP file, Keep on the download warning and then Open file.A Remote Desktop Connection window opened which I connected to with the credentials provided.

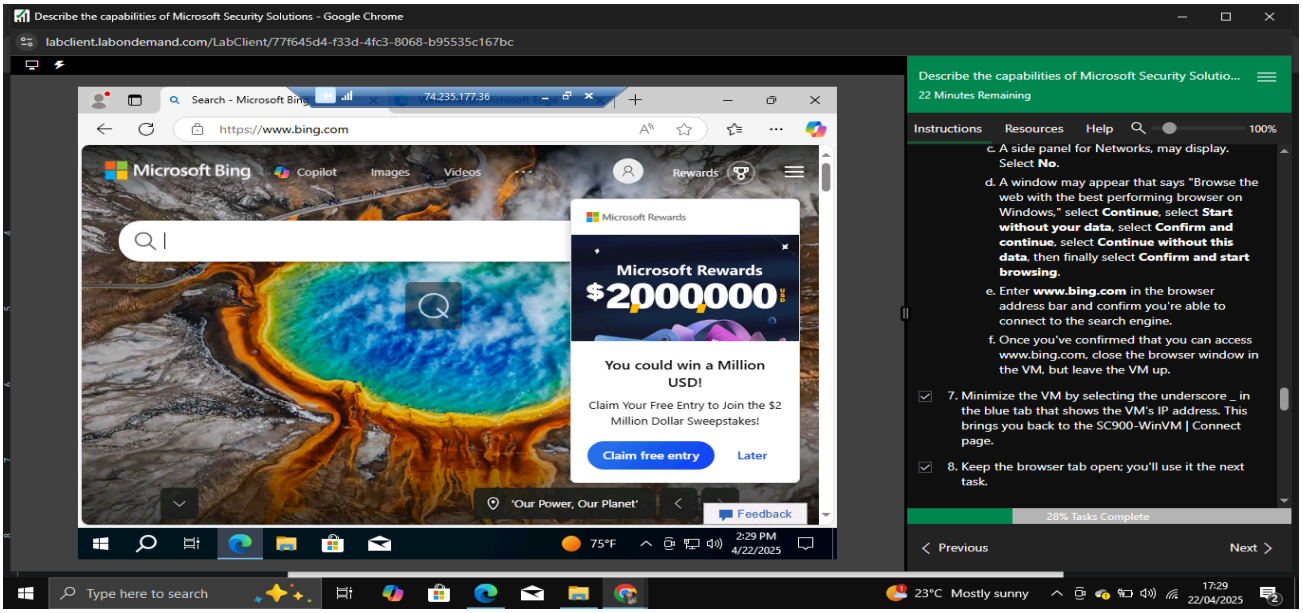
Step 5

I was able to connect to the VM because the inbound traffic rule I created allows inbound traffic to the VM via RDP. On the window to Choose privacy settings for your device,I selected Accept and No on the Network window..

Step 6

With the VM in the RDP session up and running, I tested outbound connectivity to the Internet from the VM.From the open VM,I selected Microsoft Edge to open the browser and confirmed I was able to connect to the search engine and access the website entered in the address bar.

Here is my screenshot



Step 7

I minimized the VM by selecting the underscore _ in the blue tab that shows the VM's IP address. This brought me back to the SC900-WinVM | Connect page.

Step 8

I kept the browser tab open to use it the next task

TASK 4

In this task, I went through the process of creating a custom outbound rule to block outgoing Internet traffic and test that rule.

Step 1

On the SC900-WinVM | Connect page. From the left navigation panel,I selected Networking

Step 2

I selected the Outbound port rules tab and the default rule "AllowInternetOutBound". This rule allows all outbound Internet traffic. I cannot remove the default rule, but I can override it by creating a rule with higher priority. From the right side of the page, I selected +Create port rule then from the drop-down list, Outbound port rule.

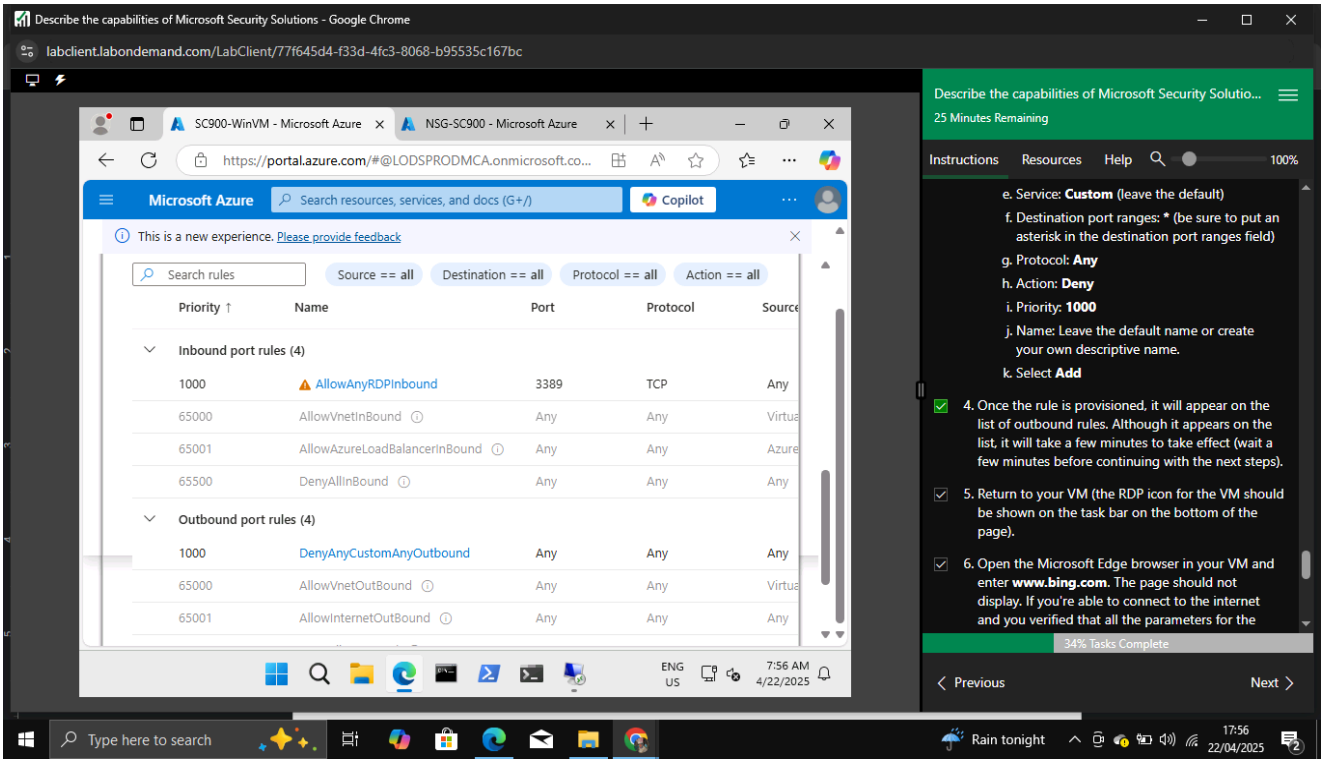
Step 3

On the Add outbound security rule page, I specified the settings provided

Step 4

Once the rule was provisioned, it appeared on the list of outbound rules

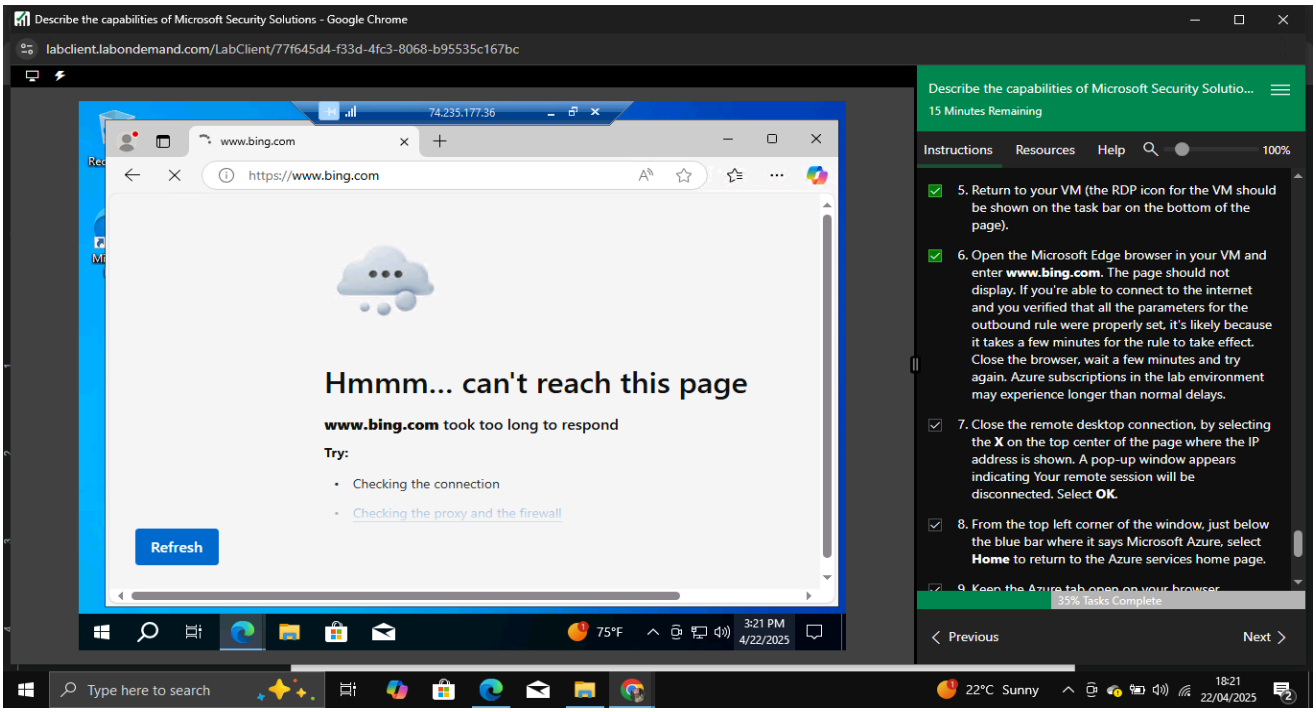
Here is my screenshot



Step 5 and 6

I returned to my VM, opened the Microsoft Edge browser and entered www.bing.com. The page should not display.

Here is my screenshot



Step 7,8 and 9

I closed the remote desktop connection,From the top left corner of the window selected Home to return to the Azure services home page and kept the Azure tab open on my browser.

1.4 LAB 3:EXPLORE MICROSOFT DEFENDER FOR CLOUD

LAB SCENARIO

In this lab,I explored Microsoft Defender for Cloud.

TASK 1

In this task,I did a high-level walk-through of some of the capabilities of Microsoft Defender for Cloud.

Step 1

I navigated to the home page for Azure services

Step 2

In the blue search bar,I entered Microsoft Defender for Cloud, then from the results list, selected Microsoft Defender for Cloud.

Step 3

I landed on the Getting started page, and was prompted to upgrade. I scrolled to the bottom of the page and selected “Remind me later”and was taken to the Overview page.

Step 4

The Overview page displayed 0 assessed resources and active recommendations initially. After refreshing the browser, the top of the page showed details such as the number of Azure subscriptions, assessed resources, active recommendations, and any security alerts. The main section featured cards for Security Posture, Regulatory Compliance, Insights, and more. The Microsoft Defender for Cloud default policy initiative had already been assigned automatically through the Azure subscription setup. The Secure Score appeared as 0% due to a possible delay of up to 24 hours for the score to update.

Step 5

I selected Assessed resources on top of the page, it took me to the inventory page that listed the current resources,I selected the virtual machine resource, sc900-winvm, and was taken to the Resource health page for the VM that provided a list of recommendations.From the available list, I selected any item from the list that showed an unhealthy status,then the drop-down arrow next to Remediation steps.I noted how remediation instructions were provided along with the option to take action, exited the window without taking any action and returned to the Microsoft Defender for Cloud overview page.

Step 6

From the left navigation panel, I selected Recommendations,Verified that All recommendations tab was selected,the dashboard view showed Active recommendations by severity, Resource health, and more,From the list, I selected an item. In the page that opened, I saw a description and additional information that included remediation steps, affected resources, and more then exited the page.

Step 7

From the main left navigation panel,I expanded Cloud Security then selected Regulatory compliance.The regulatory compliance page provides a list of compliance controls based on the Microsoft cloud security benchmark.I saw that there was no subscription to calculate compliance for, because there may be up to a 24 hour delay for information to appear.

TASK 2

In this task, I discovered how to enable/disable the various Microsoft Defender for Cloud plans.

Step 1

From the Microsoft Defender for Cloud overview page, I expanded Management and selected Environment settings from the left navigation panel.

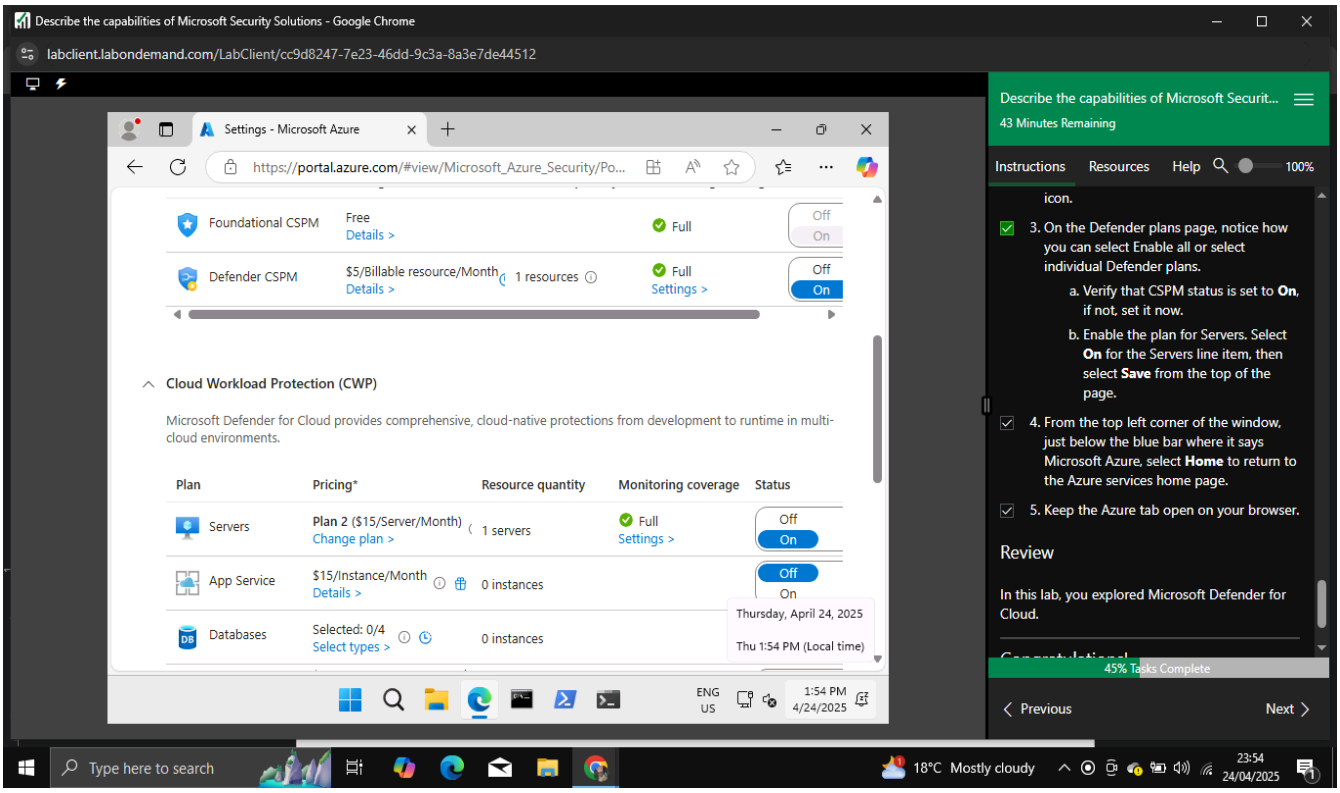
Step 2

I selected the Expand all box then MOC Subscription--lodXXXXXXXX subscription listed next to the yellow key icon.

Step 3

On the Defender plans page,I noticed how one can select Enable all or select individual Defender plans, I verified that CSPM status was set on and set it,enabled the plan for servers by selecting On for the Servers line item, then selected Save from the top of the page.

Here is my screenshot



Step 4 and 5

I selected Home to return to the Azure services home page and kept the tab open.

1.5 LAB 4:EXPLORE MICROSOFT SENTINEL

LAB SCENARIO

In this lab, I walked through the process of creating a Microsoft Sentinel instance, set up the permissions to ensure access to the resources that will get deployed to support Microsoft Sentinel,walked through the steps for connecting Microsoft Sentinel to data sources, set up a workbook, and did a brief walk-through of some of key capabilities available in Microsoft Sentinel.

TASK 1

I created a Microsoft Sentinel instance.

Step 1

I navigated to the home page for Azure services.

Step 2

In the blue search box on the top of the page, I entered Microsoft Sentinel then selected Microsoft Sentinel from the search results.

Step 3

From the Microsoft Sentinel page, I selected Create Microsoft Sentinel.

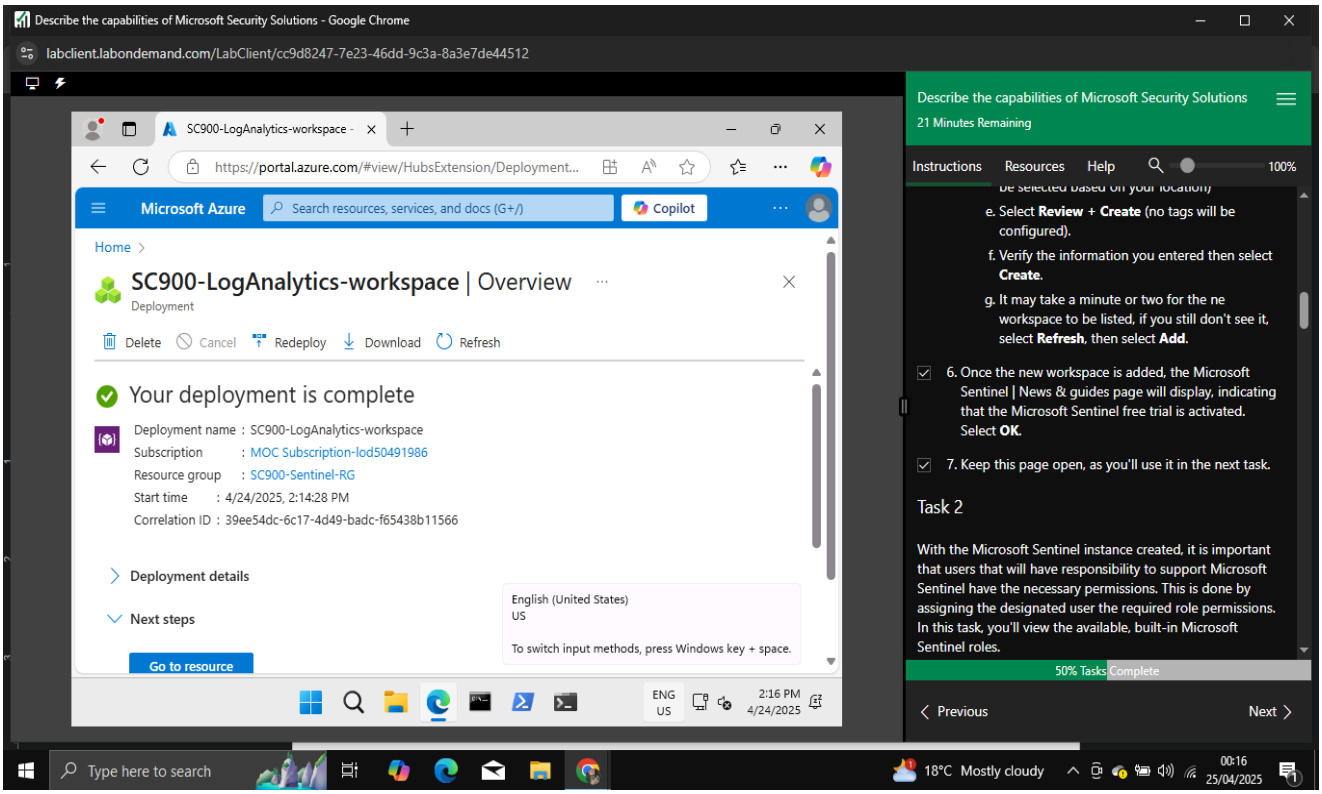
Step 4

From the Add Microsoft Sentinel to a workspace page, selected Create a new workspace

Step 5

In the basics tab of the Create Log Analytics workspace, I entered the information provided in the instructions to create my workspace.

Here is my screenshot



Step 6

Once the new workspace was added, the Microsoft Sentinel | News & guides page was displayed, indicating that the Microsoft Sentinel free trial was activated. I selected OK.

Step 7

I kept the page in the above task open as I would use it in the next task.

TASK 2

In this task,I viewed the available, built-in Microsoft Sentinel roles.

Step 1

In the blue search box, I entered resource groups then selected Resource groups from the search results.

Step 2

From the Resource groups page, I select the resource group that I Created with Microsoft Sentinel,SC900-Sentinel-RG.

Step 3

From the SC900-Sentinel-RG page,I selected Access control (IAM) from the left navigation panel

Step 4

From the Access control page, I selected View my access. For the Azure subscription provided by the Authorized Lab Hoster, a role had been defined that would give access to manage all necessary resources.

Step 5

From the Access control page, I selected the Roles tab on the top of the page,entered Microsoft Sentinel in the search box to view the built-in roles associated with Microsoft Sentinel.From any of the roles listed, selected view to view the details of that role.I understood that as a best practice should assign the least privilege required for the role.

Step 6

I closed the window on access control page

Step 7 and 8

I returned to the Azure services home page and kept the azure portal open on my browser.

TASK 3

In this step, I used the content hub to deploy the Microsoft Defender for Cloud solution for Microsoft Sentinel. This solution allows to ingest Security alerts reported in Microsoft Defender for Cloud.

Step 1

From the Azure services home page,I selected Microsoft Sentinel, then the instance I created, SC900-LogAnalytics-workspace.

Step 2

From the left navigation panel,I expanded Content Management then selected Content hub.

Step 3

I selected Microsoft Defender for Cloud from the list of solutions,from the side window that opened,read the description then selected Install.Once the installation was completed, the status column in the main window showed installed.

Step 4

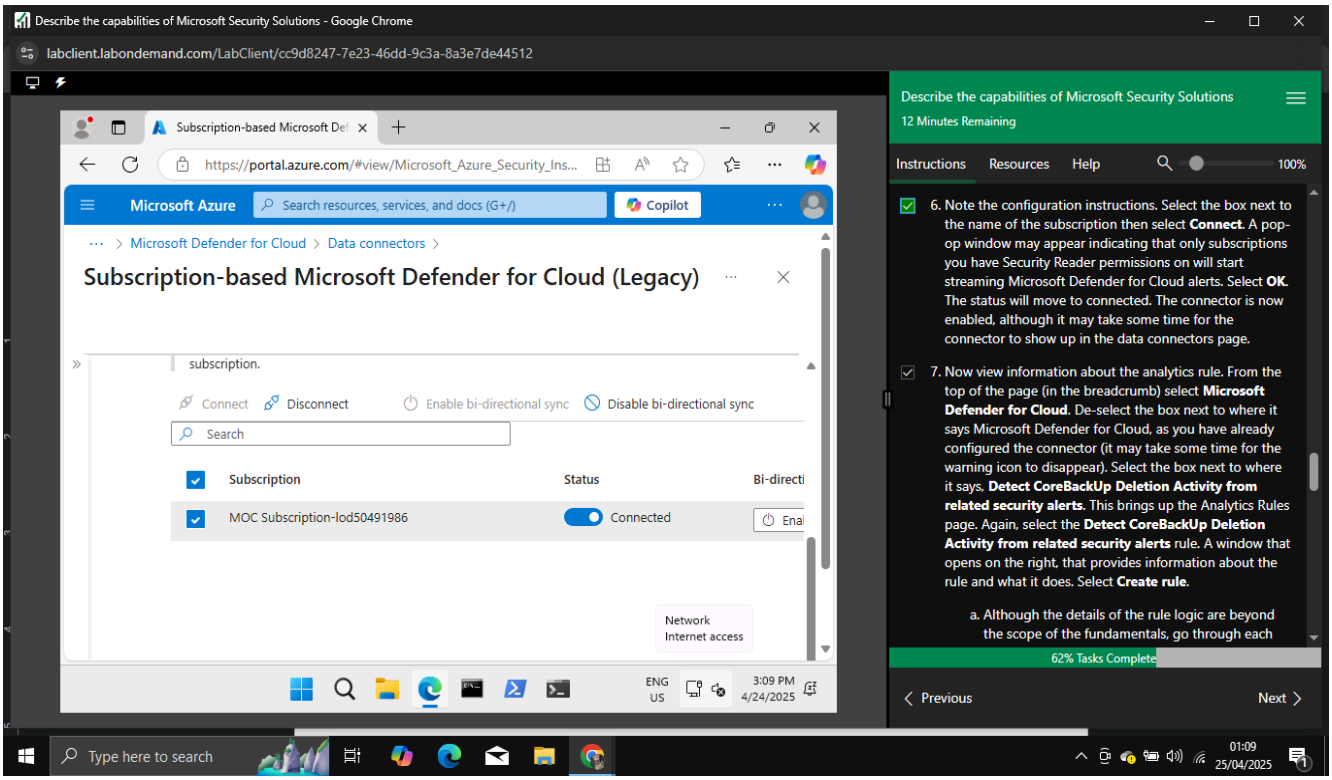
Once again, I selected Microsoft Defender for Cloud from the list and manage from the window on the right.

Step 5

On the right side of the Microsoft Defender for Cloud page was the description and notes associated with the solution from Content Hub and what was included as part of that solution.I selected the box next to where it says Subscription-based Microsoft Defender for Cloud (Legacy). A window opened on the right side of the page where I then selected the Open connector page.

Step 6

I noted the configuration instructions. Selected the box next to the name of the subscription then Connect. The status was moved to connected.Here is my screenshot



Step 7

I viewed the information about the analytics rule.I selected the Detect CoreBackUp Deletion Activity from the related security alerts rule. A window opened on the right, that provided information about the rule and what it does,I selected Create rule, Review + create tab and saved.

Step 8

ON the sentinel page I selected Content hub from the bread-crumb at the top of the page, above the Analytics rules.

Step 9

I kept the page from step 8 open, as I would use it in the next task.

TASK 4

In this task,I walked through some of the options available in Sentinel.

Step 1

From the left navigation panel, I expanded Threat management and explored the options listed.

Step 2

From the left navigation panel, I expanded Content Management, then selected Community.The community page included Cybersecurity insights and updates from Microsoft Research.

Step 3

From the left navigation panel, I expanded on Configuration and explored the options listed.Under automation I saw how to create simple automation rules.

Step 4 and 5

I closed my current window and selected Microsoft Azure to return to the home page of the Azure portal.

Step 6

I signed out and closed all the open browser tabs.

1.6 LAB 5:EXPLORE MICROSOFT DEFENDER FOR CLOUD APPS

LAB SCENARIO

In this lab,I explored the capabilities of Microsoft Defender for Cloud Apps.Walked through the information available on the Cloud Discovery dashboard, the Cloud app catalog, capabilities available to investigate findings, and ways to control impact to organization through policies.

TASK 1

In this task,I explored Cloud Discovery

Step 1

I opened Microsoft Edge and entered the url provided in the address bar.

Step 2

I Signed in with my admin credentials for the Microsoft 365 tenant.

Step 3

From the left navigation pane of the Microsoft 365 admin center, I selected Show all.

Step 4

I selected security under admin centers

Step 5 and 6

I was taken to the Microsoft 365 defender portal,from the left navigation panel, selected Cloud apps to expand the list then Cloud Discovery. This took me to the Dashboard view.

Step 7

I selected Discovered apps which provided a more detailed view of the discovered apps, including risk score, traffic, number of users and more.

Step 8

I selected Actions on the top-right corner of the page,Create Cloud Discovery snapshot report,and Next. Here one would fill in the requested details and upload traffic logs to generate and upload a report.I also explored the option for continuous reports.

Step 9

From the top-right corner of the screen, I selected Actions then Cloud Discovery Settings.From the left side of the screen, under Connected apps, selected App connectors.On the Connected apps page I explored the office 365 and returned to the cloud discovery dashboard.

TASK 2

In this task I explored the capabilities of the Cloud app catalog.

Step 1

From the left navigation panel, I selected Cloud app catalog

Step 2

I selected Compliance risk factor to view the available options that can further be filtered by risk score, by moving the sliders on the risk score on the top of the page.

Step 3

Admins can also search for apps by category. For example, in the search for category field, I entered Social network, then selected Social network.Hovering the mouse over any topics for a given category shows an information icon that one can select to get more information about that topic.

Step 4

I kept this page in step 3 above open as I would use it in my next task.

TASK 3

In this lab,I explored ways in which I can investigate the recorded activities with the activity log and files.

Step 1

From the left navigation panel, I selected Activity Log.

Step 2

To provide data protection, Microsoft Defender for Cloud Apps gives visibility into all the files from your connected apps, for example all the files stored in SharePoint and Salesforce. From the left navigation pane, I selected and explore the Files option

Step 3

I kept this page in step 3 above open as I would use it in my next task.

TASK 4

In this task, I explored the policies in Microsoft Defender for Cloud Apps.

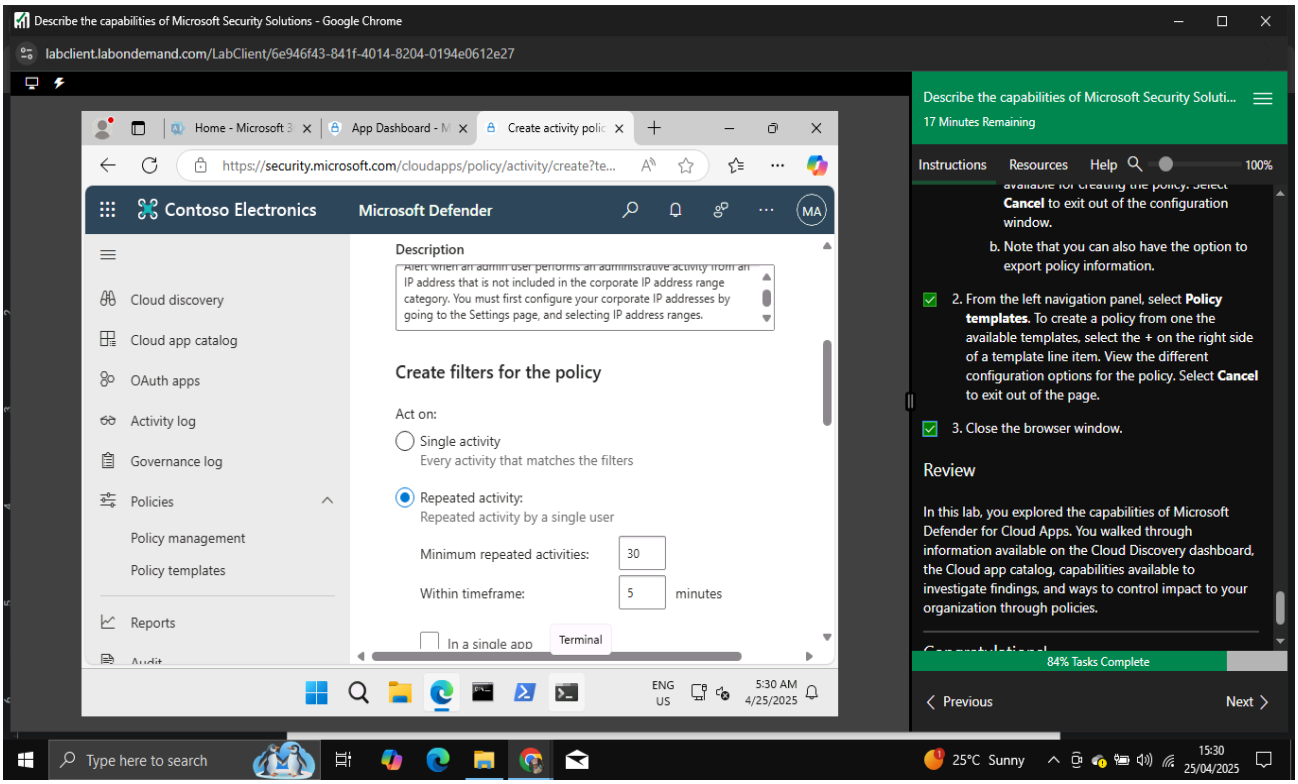
Step 1

From the left navigation panel, I selected Policies then Policy management and explored the options there..

Step 2

From the left navigation panel, I selected Policy templates.To create a policy from one of the available templates, I am to select the + on the right side of a template line item.I viewed the different configuration options for the policy,selected cancel to exited out of the page.

Here is my screenshot



Step 3

I closed the browser window

1.7 EXPLORE THE MICROSOFT DEFENDER PORTAL

LAB SCENARIO

In this lab,I explored the Microsoft Defender portal by walking through the content displayed on the landing page,the options on the navigation panel that provide quick access to functionality that is part of Microsoft's Extended Detection and Response (XDR) solution: Microsoft Defender for Endpoints, and Microsoft Defender for Office 365 (email and collaboration) and how Microsoft Secure Score can help an organization improve its security posture.

TASK 1

In this task, I explored the Microsoft Defender landing page.

Step 1

I opened Microsoft Edge and entered admin.microsoft.com in the address bar.

Step 2

I signed in with my admin credentials

Step 3

From the left navigation pane of the Microsoft 365 admin center, under Admin centers,I selected Security.

Step 4

A new browser page opens to the welcome page of the Microsoft Defender portal.

Step 5

I Scrolled through the page to view the default set of cards for my role as global admin.

Step 6

The cards displayed can be customized to your preference. I Selected + Add cards. A Window opened that displays any cards that were available to add to home page

Step 7

Selecting the ellipses on the top-right of any card provided the option to remove the card from the landing page.

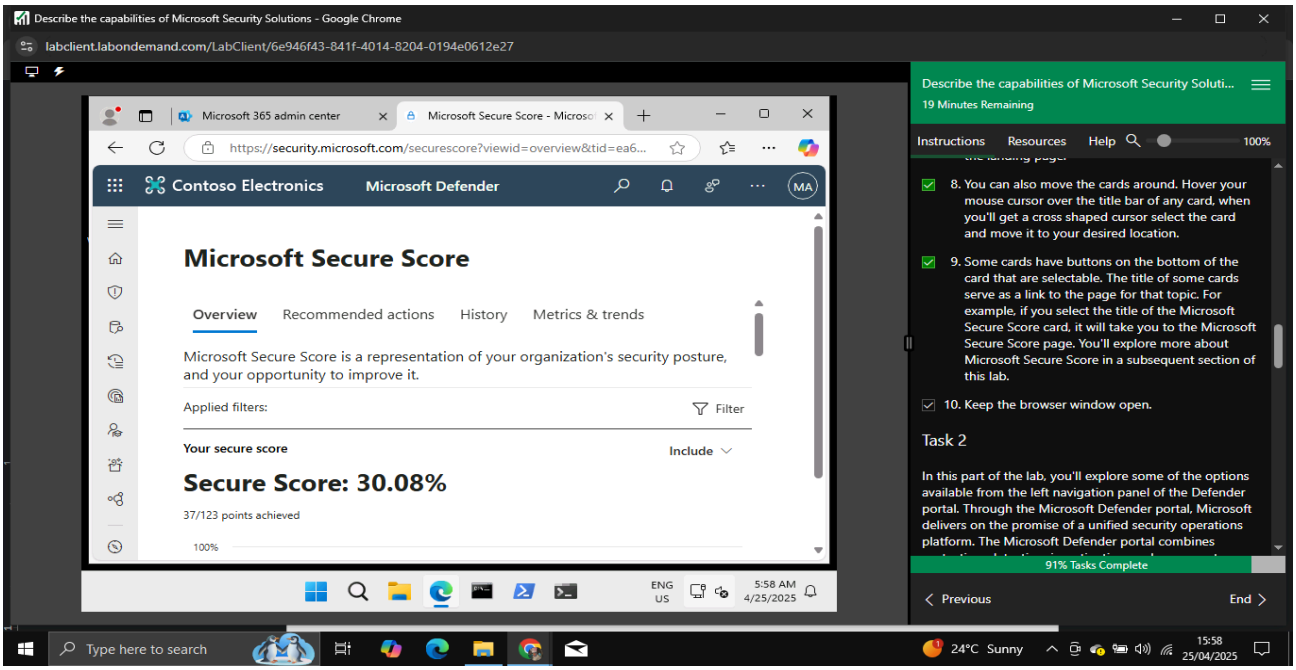
Step 8

I could also move the cards around.I Hovered my mouse cursor over the title bar of any card, got a cross shaped cursor that would enable me to select the card and move it to my desired location.

Step 9

Some cards had buttons on the bottom of the card that were selectable and served as a link to the page for that topic. I selected the title of the Microsoft Secure Score card, and it took me to the Microsoft Secure Score page.

Here is my screenshot



Step 10
I kept the browser window from step 9 open.

TASK 2

In this part of the lab, I explored some of the options available from the left navigation panel of the Defender portal.Through the Microsoft Defender portal, Microsoft delivers on the promise of a unified security operations platform. The Microsoft Defender portal combines protection, detection, investigation, and response to threats across an entire organization and all its components, in a central place.

Step 1
I explored the left navigation panel at will.

Step 2
I returned to the home page of the Microsoft Defender portal.

TASK 3

In this task, I explored how Microsoft Secure Score can help an organization improve its security posture.

Step 1
From the left navigation panel of Microsoft Defender Portal,I expanded Exposure Management than selected Secure score.

Step 2
I selected Include, next to where it says Your secure score.

Step 3
The overview page also included top improvement actions, comparison score, history, and additional resources.

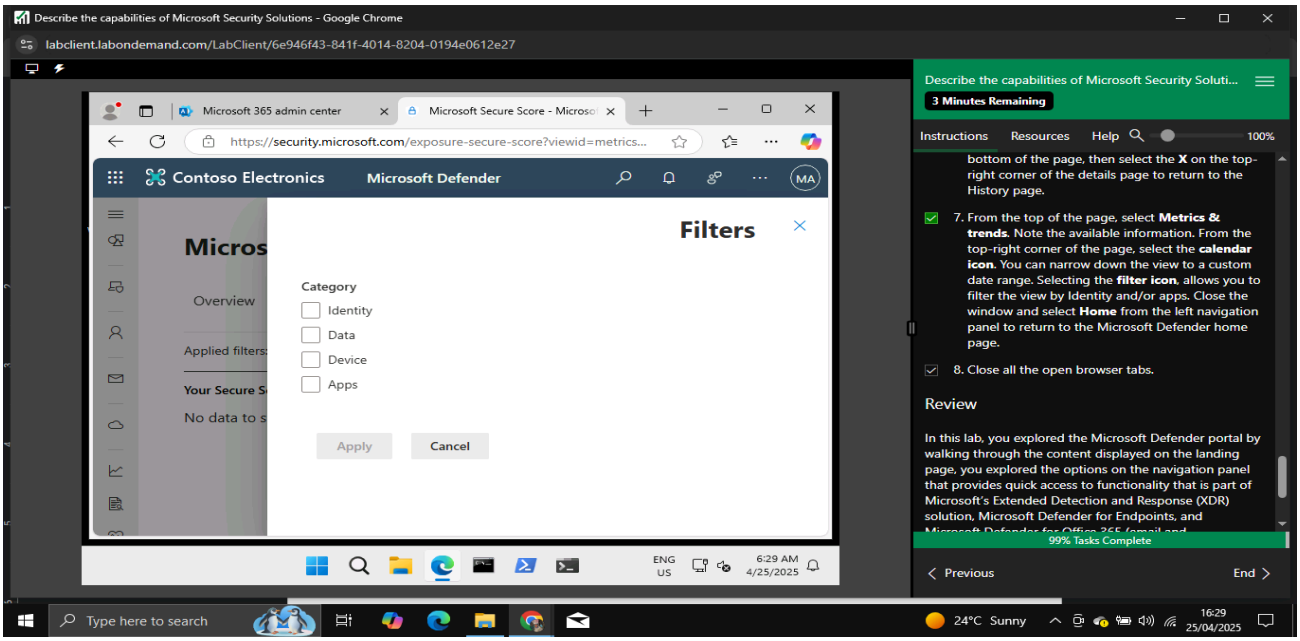
Step 4
I selected Recommended actions from the top of the page and noticed the information available in the table.

Step 5
I selected the first item from the list and reviewed the available information. In the window that opened, noted the status options available. Selected the Implementation tab to view information related to implementation and closed that window.

Step 6
I explored the History tab from the top of the page.

Step 7
I selected Metrics & trends,the calendar icon and explored the options therein.The filter icon allows to filter the view by Identity and/or app.

Here is a screenshot



Step 8

I closed all the open browser tabs and completed the lab.

1.8 CONCLUSION AND SUMMARY

Through these labs, I gained hands-on experience with Microsoft security tools, enhancing my understanding of cloud and endpoint protection. I learned how to enable audit logs, configure NSG rules, and explore threat detection features in Microsoft Defender for Cloud. Setting up Microsoft Sentinel and connecting it to data sources demonstrated the power of centralized security monitoring. I also gained insights into managing cloud app risks and improving security posture using Defender for Cloud Apps and Secure Score. Overall, the labs offered a practical foundation for securing modern IT environments using Microsoft technologies.